

Date: 02 July 2026	Team ID: SB-7429
Project Name: Credit Card Approval Prediction using Machine Learning	Maximum Marks: 2 Marks

TECHNOLOGY STACK TEMPLATE

Define Your Technology Stack

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution.

S.No	Architecture Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client UI	HTML5, CSS3, JavaScript (ES6)	Vanilla web tools enable rapid rendering, responsiveness across devices, and clear Dark Fintech styling without dependencies.
2	Backend Web Server	Python, Flask (2.0+)	Python-native micro-framework. Extremely lightweight, exposes RESTful endpoints and integrates with model pickles easily.

Technology Stack Details (Continued)

S.No	Architecture Layer	Technology Chosen	Justification / Purpose
3	Machine Learning & Data	Pandas, NumPy, Scikit-Learn, Imbalanced-Learn	Industry-standard libraries for tabular data preprocessing, model fitting (Random Forest), scaling, and SMOTE balancing.
4	Database & Model Storage	Pickle Format & Local CSV Cache	High-speed binary serialization of model configurations, StandardScaler weights, and calibrated threshold floats.

Technology Stack Details (Continued)

S.No	Architecture Layer	Technology Chosen	Justification / Purpose
5	Version Control	Git & GitHub	Facilitates seamless team collaboration, branch management, pull-request reviews, and historical code tracking.
6	Cloud Deployment Hosting	Render Cloud Platform	Free-tier web service deployment, automated Git integration, and robust Python runtime execution environment.